

**EQUIPMENT & PROJECT SUPPORT**

# Pumping systems. Built around your well.

Electric submersible pumping equipment for geothermal energy and oil & gas production.



## Equipment. Selection. Support.

A compact guide to the equipment portfolio, selected technical reference data and the information needed to start your project.

[office@repnfzco.com](mailto:office@repnfzco.com)

## 01 / PORTFOLIO

# Two applications. One clear portfolio.

Start with the application, then select the equipment family. Configuration and service scope are agreed for the project.

| Equipment family             | Geothermal | Oil & Gas |
|------------------------------|------------|-----------|
| ESP pumps                    | Yes        | Yes       |
| Submersible motors           | Yes        | Yes       |
| Protectors & seal sections   | Yes        | Yes       |
| Gas handling                 | —          | Yes       |
| Intake sections              | —          | Yes       |
| Cable systems                | Yes        | Yes       |
| Downhole monitoring          | Yes        | —         |
| Variable speed drives        | Yes        | —         |
| Corrosion-resistant coatings | —          | Yes       |

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## 02 / GEOTHERMAL PUMPS

# Four pump families.

Single-stage reference operating points. TP diffusers are assembled inside a housing; QJ uses bolted diffusers without a continuous outer housing.

| Series  | Hz | rpm  | Flow<br>m <sup>3</sup> /h | Head<br>m/stage | Power<br>kW/stage | Range m <sup>3</sup> /h |
|---------|----|------|---------------------------|-----------------|-------------------|-------------------------|
| TP 677  | 50 | 2910 | 165                       | 15.9            | 12.9              | 125-208                 |
| TP 677  | 60 | 3500 | 200                       | 23              | 22.4              | 150-250                 |
| TP 740* | 50 | 2910 | 250                       | 11.5            | 10.7              | 170-290                 |
| TP 740  | 60 | 3500 | 300                       | 16.6            | 18.7              | 210-360                 |
| TP 905  | 50 | 2910 | 405                       | 26.3            | 30.1              | 330-520                 |
| TP 905  | 60 | 3500 | 490                       | 38              | 52.3              | 400-625                 |
| QJ 1047 | 50 | 2910 | 350                       | 38.8            | 51.6              | 208-455                 |
| QJ 1047 | 60 | 3500 | 425                       | 56.2            | 89.8              | 250-550                 |

\* TP 740 at 50 Hz: this operating point is pending confirmation. Request confirmed performance data before equipment selection.

## Construction and materials

| Item                       | Selected options                                                 |
|----------------------------|------------------------------------------------------------------|
| Pump outside diameter      | TP 677: 172 mm; TP 740: 188 mm; TP 905: 230 mm; QJ 1047: 266 mm  |
| Stage assembly             | FLT floater; PKT packet; CPM compression                         |
| Stages                     | Ni-resist cast iron; stainless powder-alloy option               |
| Shaft                      | High-strength corrosion-resistant steel; Alloy 925 (718) options |
| Radial bearings            | VK8-V hard alloy                                                 |
| Abrasion-resistant options | AR1: up to 500 mg/L solids; AR2: up to 1,000 mg/L solids         |

Head and power values above apply to one stage at the listed operating point. They are not the total head or power of a complete multistage pump.

### 03 / GEOTHERMAL MOTORS

# 728-series permanent magnet motors.

Housing outside diameter: 185 mm. Model codes below retain their catalogue designation.  
Power and voltage depend on rated speed.

| Model      | kW 3000 / 3600 | V 3000 / 3600 | A   | Length mm | Weight kg | Min. flow m/s |
|------------|----------------|---------------|-----|-----------|-----------|---------------|
| 728-326HP  | 200 / 240      | 2800 / 3360   | 45  | 3937      | 592       | 0.20          |
| 728-408HP  | 250 / 300      | 3300 / 3960   | 48  | 4353      | 665       | 0.25          |
| 728-489HP  | 300 / 360      | 3100 / 3720   | 61  | 4769      | 738       | 0.30          |
| 728-570HP  | 350 / 420      | 3500 / 4200   | 63  | 5185      | 811       | 0.40          |
| 728-652HP  | 400 / 480      | 3100 / 3720   | 81  | 5601      | 884       | 0.40          |
| 728-734HP  | 450 / 540      | 3500 / 4200   | 81  | 6017      | 958       | 0.40          |
| 728-815HP  | 500 / 600      | 3300 / 3960   | 97  | 6017      | 958       | 0.45          |
| 728-897HP  | 550 / 660      | 3800 / 4560   | 92  | 6433      | 1030      | 0.45          |
| 728-978HP  | 600 / 720      | 3900 / 4680   | 98  | 7265      | 1176      | 0.45          |
| 728-1060HP | 650 / 780      | 4500 / 5400   | 90  | 7681      | 1249      | 0.50          |
| 728-1142HP | 700 / 840      | 3800 / 4560   | 117 | 8097      | 1322      | 0.50          |
| 728-1224HP | 750 / 900      | 4000 / 4800   | 119 | 8513      | 1395      | 0.55          |
| 728-1305HP | 800 / 960      | 4700 / 5640   | 106 | 8929      | 1468      | 0.60          |

## Thermal configuration

| Standard     | HT           | UHT          |
|--------------|--------------|--------------|
| 120 °C fluid | 160 °C fluid | 200 °C fluid |

Select the motor rating, fluid-temperature class and cooling-flow velocity together. Corrosion-resistant options include coated housings and stainless-steel end parts, depending on the selected model configuration.

## 04 / OIL & GAS

# Pumps and downhole drives.

Modular centrifugal pumps and submersible motors are selected together with the protector, power cable and surface drive.

## Oilfield pump series

| Series | Housing outside diameter | Stage assembly |
|--------|--------------------------|----------------|
| 362    | 92 mm / 3.62 in          | FLT / PKT      |
| 400    | 103 mm / 4.06 in         | FLT / PKT      |
| 512    | 130 mm / 5.12 in         | FLT / PKT      |

The reference oilfield portfolio spans approximately 315-3,144 BPD. Final pump model, stage count, materials and operating range are confirmed against the well conditions.

## Performance data for your duty

Request the head, power and efficiency curves for the selected pump model and operating speed. Include flow, required head, fluid temperature, gas content and solids in your enquiry.

## Induction or permanent magnet

| Drive type       | Configuration                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Induction        | Three-phase, oil-filled, two-pole squirrel-cage motor for continuous S1 service. The reference family lists a synchronous-speed range of 2,100-4,200 rpm. |
| Permanent magnet | Submersible permanent-magnet motor selected with a compatible variable speed drive and the required operating-speed range.                                |

For induction motors, synchronous speed is the rotating-field speed. Actual shaft speed depends on slip and load. Model-specific electrical ratings and dimensions are provided with the motor selection.

## Provide these selection inputs

Pump shaft load and speed; casing diameter and installation geometry; fluid temperature and cooling flow; available voltage and drive; cable length; protector interface and monitoring requirements.

# Power cables by application

Select the cable construction together with the motor and well conditions.

## Geothermal | REPN-FX / REPN-QX

| Parameter            | REPN-FX                          | REPN-QX                          |
|----------------------|----------------------------------|----------------------------------|
| Rated voltage        | 5 kV                             | 5 kV                             |
| Conductors           | 3 × 33.5 mm <sup>2</sup> / 2 AWG | 3 × 33.5 mm <sup>2</sup> / 2 AWG |
| Cable designation    | REPN-FX-5KV                      | REPN-QX-5KV                      |
| Temperature rating   | 232 °C                           | 204 °C                           |
| Insulation           | Polyimide film + FEP             | Polyimide film + EPDM            |
| Core sheath          | FEP                              | Lead                             |
| Armour               | Stainless steel tape             | Stainless steel tape             |
| Maximum overall size | 14.1 × 35.7 mm                   | 17.4 × 44.5 mm                   |

## Oil & Gas | EPR / lead reference construction

| Parameter                 | Specification                                               |
|---------------------------|-------------------------------------------------------------|
| Rated voltage / frequency | 5 kV AC / 50 Hz                                             |
| Conductor sizes           | 3 × 10 / 13.3 / 16 / 21.15 / 25 / 33.6 / 35 mm <sup>2</sup> |
| Conductor temperature     | Maximum continuous 230 °C                                   |
| Insulation / core sheath  | EPR / lead                                                  |
| Armour options            | Galvanized steel; corrosion-resistant steel or Monel alloy  |
| Hydrostatic pressure      | Maximum 40 MPa for this reference construction              |

Detailed dimensions, fluid limits and oilfield current ratings are provided in the ESP cable data sheet. The ratings of different cable constructions are not interchangeable.

## 06 / DOWNHOLE ASSEMBLY

# Protection and fluid handling.

Match each component to the pump, motor, fluid and operating range.

## Protectors and seal sections

| Application | Configuration overview                                                                                                                          |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Geothermal  | Tandem labyrinth and diaphragm arrangements. Housing OD 172 mm; shaft diameter 43 mm; oil capacity 30 L. Axial-load options: 2,300 / 3,200 kgf. |
| Oil & Gas   | Modular single-housing and tandem arrangements, with standard, reinforced or tilting-pad thrust-bearing options.                                |
| Selection   | Confirm chamber sequence, thermal configuration, shaft and connection dimensions, and thrust capacity together with the drive assembly.         |

## Gas separation and dispersion

Hydrocyclone separators remove free gas ahead of the pump. Dispersing stages produce a more homogeneous gas-liquid mixture. Combined modules provide both functions.

| Selected module                           | Intake gas content | Separation ratio |
|-------------------------------------------|--------------------|------------------|
| Hydrocyclone separator                    | ≤ 70%              | ≤ 87%            |
| Separator-dispersant                      | ≤ 75%              | ≤ 87%            |
| Upgraded dispersing unit                  | ≤ 80%              | ≤ 87%            |
| Double upgraded unit / dynamic gas outlet | ≤ 90%              | ≤ 90%            |

Values apply only to the described module configurations. Selection depends on gas content at the intake and the full fluid properties.

## Intakes and surface protection

Intake modules guide formation fluid into the pump. Material options include corrosion-resistant housings and hard-alloy radial bearings. Selected CR2C seal-section configurations use HVOF metallic coating with a minimum thickness of 150 µm.

**07 / CONTROL & SUPPORT**

# From equipment to the project.

Surface equipment, downhole monitoring and service scope are defined around the selected pumping system.

## Geothermal variable speed drives

| Parameter                 | Reference specification       |
|---------------------------|-------------------------------|
| Input voltage / frequency | 380 / 480 V AC; 47-63 Hz      |
| Output frequency          | 5-500 Hz; 0.1 Hz resolution   |
| Short-term overload       | 120% for 1 minute             |
| Communications            | RS485 / Modbus, SCADA and I/O |
| Filtering                 | Sine-wave filter              |

## Downhole monitoring

The geothermal configuration includes environment temperature and pressure and motor-oil temperature. Pump-outlet temperature, pressure and flow are optional measurements, subject to the selected sensor and surface system.

## Project and service support

Engineering and equipment selection · Installation and commissioning support · Monitoring and troubleshooting · Inspection and failure analysis · Repair and testing · Replacement and optimization.

Service scope is agreed for the individual project. Share equipment details, operating history and the outcome you need.

**Discuss your application.**

**[office@repnfzco.com](mailto:office@repnfzco.com)**

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